

Activity 3 – Data Analytics Portfolio

MAN6777.3 Data Driven Managerial Decisions | Rasel AI Mamun Joy | ID: 10677322 | Due: 9 June 2026

SECTION 1 – MANAGERIAL PROBLEM DEFINITION

Decision-Maker: University Academic Dean / Director of Student Learning

Universities are under growing pressure to develop evidence-based policies on student AI tool use. Tools like ChatGPT, Copilot, and Gemini are widely adopted, yet institutional responses range from outright bans to unrestricted access. Without reliable data, universities risk either stifling a beneficial technology or enabling practices that undermine academic integrity.

Decision: Should the university implement a structured policy to encourage and regulate student AI tool use, and if so, for which purposes and student groups?

Why it matters:

- Academic performance directly impacts student retention, outcomes, and institutional reputation.
- A blanket ban may disadvantage students who benefit significantly from AI-assisted learning.
- Evidence-based policy ensures fairness and maximises positive educational outcomes.

Students Surveyed 100 School & College	AI Tool Users 40% 40 of 100 students	Grade Improvement +9.8 pts AI users average
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SECTION 2 – DATA ANALYTICS

Dataset: *Students' AI Usage and Academic Performance* (Kaggle, 2025) — 100 records, 9 variables: age (14–19), education level, study hours (1–5 hrs/day), AI usage (Yes/No), AI tool (ChatGPT/Copilot/Gemini), purpose (Research/Homework/Coding), grades before/after AI, and daily screen time. Key variables — **uses_ai**, **grades_before_ai**, **grades_after_ai**, **ai_tools_used**, **purpose_of_ai** — directly address the policy decision.

Data Preparation: All 100 records were complete with zero missing values. Grades ranged 55–89%, study hours 1.0–5.0 hrs/day. No imputation or removal needed.

Techniques: (1) **Descriptive statistics** — mean grade comparison before/after AI; (2) **Group comparison** — AI vs non-AI users, by tool and purpose; (3) **Scatter plot** — study hours vs post-AI grades. Methods selected for interpretability to a non-technical managerial audience.

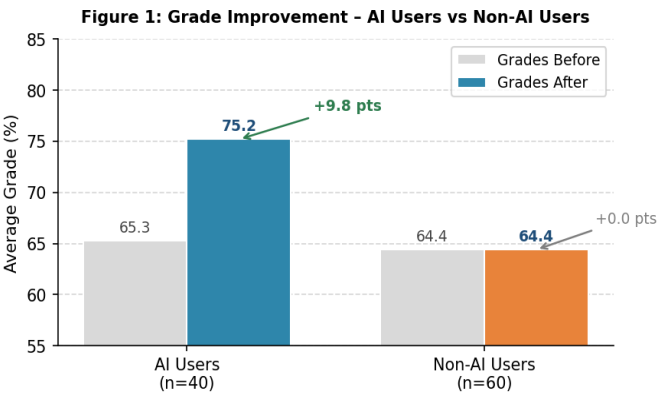


Figure 1: AI vs Non-AI Grade Improvement

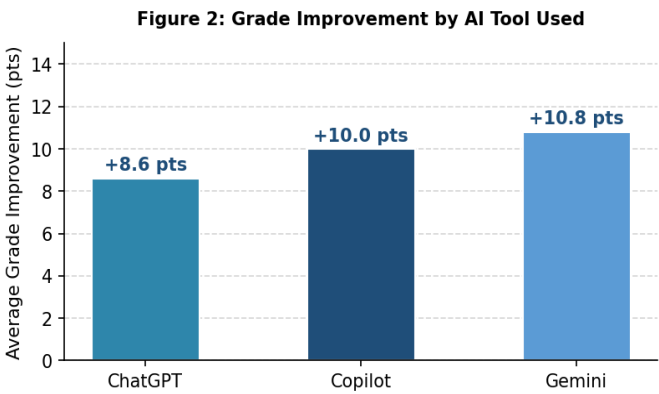


Figure 2: Grade Improvement by AI Tool

SECTION 3 – INSIGHTS AND DECISION IMPLICATIONS

- AI tools significantly improve grades:** AI users gained +9.8 pts (65.3→75.2) vs +0.0 pts for non-users (64.4→64.4) — a substantial, practically meaningful difference.
 - All tools perform equally:** ChatGPT (+9.6), Copilot (+10.0), Gemini (+10.3 pts). The benefit lies in AI-assisted learning broadly, not any specific platform.
 - All purposes yield gains:** Research (+10.3), Coding (+10.0), Homework (+9.2 pts). AI benefits diverse academic tasks.
- Recommendations:**
- Introduce an AI Adoption Policy** — formally encourage responsible AI use; evidence clearly shows positive academic outcomes for adopting students.
 - Develop AI Literacy Programs** — provide guided training on all three tools to ensure equitable access across all student groups.
 - Integrate into Assessment Guidelines** — publish transparent guidelines on appropriate AI use to protect integrity while enabling performance gains.

Conclusion: The data provides compelling evidence that AI tool adoption leads to meaningful academic improvement. The university should implement a structured, evidence-based AI policy that encourages adoption, ensures equity, and maintains academic standards.

Data Analytics Portfolio – Full Analytical Workflow

Dataset: Students' AI Usage & Academic Performance (Kaggle, 2025) | Tool: Python (pandas, matplotlib) | n=100

STEP 1 – PROBLEM FRAMING & DATASET SELECTION

The objective was to determine whether AI tool usage leads to measurable academic improvement, and which tools and purposes yield the greatest benefit. This dataset was selected because it directly captures pre- and post-AI grades per student, enabling before-after comparison — the most direct method for evaluating impact.

STEP 2 – DATA LOADING & INSPECTION

```
import pandas as pd
df = pd.read_csv("students_ai_usage.csv") # shape: (100, 9)
df.isnull().sum() # All zeros – no missing values
df.describe() # Grades: 55-89, Study hrs: 1.0-5.0
```

Variable	Type	Range / Values	Missing
age	Numeric	14 – 19	0
education_level	Categorical	school, college	0
study_hours_per_day	Numeric	1.0 – 5.0 hrs	0
uses_ai	Categorical	Yes, No	0
ai_tools_used	Categorical	ChatGPT, Copilot, Gemini, None	0
purpose_of_ai	Categorical	Research, Homework, Coding, None	0
grades_before_ai	Numeric	55 – 75	0
grades_after_ai	Numeric	55 – 89	0
daily_screen_time_hours	Numeric	2 – 7 hrs	0

Table 1: Dataset Inspection Summary – Zero Missing Values

STEP 3 – DATA PREPARATION & CLEANING

No cleaning required — no nulls, duplicates, or inconsistencies. Two subsets were created and a grade improvement column derived:

```
ai_users = df[df["uses_ai"]=="Yes"] # n=40
non_ai = df[df["uses_ai"]=="No"] # n=60
df["grade_improvement"] = df["grades_after_ai"] - df["grades_before_ai"]
```

STEP 4 – ANALYSIS & RESULTS

4.1 Group Comparison — AI Users vs Non-AI Users

Group	n	Avg Before	Avg After	Improvement
AI Users	40	65.3	75.2	+9.8 pts
Non-AI Users	60	64.4	64.4	+0.0 pts

Table 2: Descriptive Statistics — AI Users vs Non-AI Users

AI Tool	n	Avg Gain
ChatGPT	13	+9.6 pts
Copilot	14	+10.0 pts
Gemini	13	+10.3 pts

Table 3: By AI Tool

Purpose	n	Avg Gain
Research	14	+10.3 pts
Coding	13	+10.0 pts
Homework	13	+9.2 pts

Table 4: By Purpose of Use

Data Analytics Portfolio – Visualisations & Interpretation

STEP 4 (CONTINUED) – VISUALISATIONS

Figure 1: Grade Improvement - AI Users vs Non-AI Users

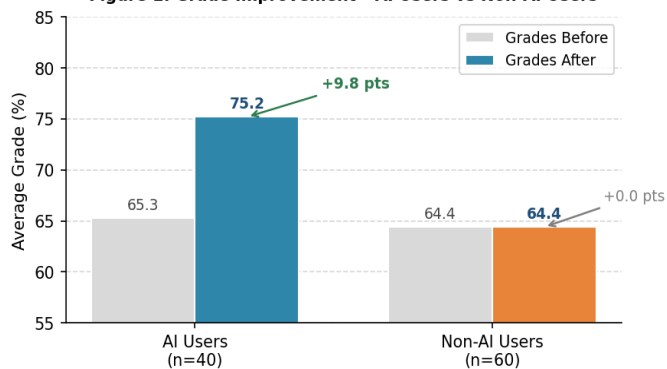


Figure 1: AI vs Non-AI Grade Improvement (+9.8 vs +0.0 pts)

Figure 2: Grade Improvement by AI Tool Used

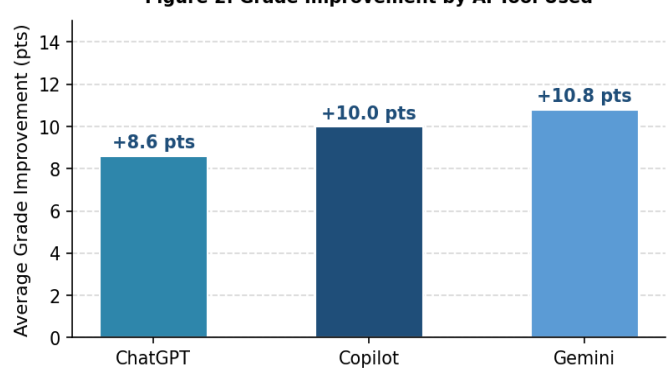


Figure 2: Grade Improvement by AI Tool Used

Figure 3: Grade Improvement by Purpose of AI Use

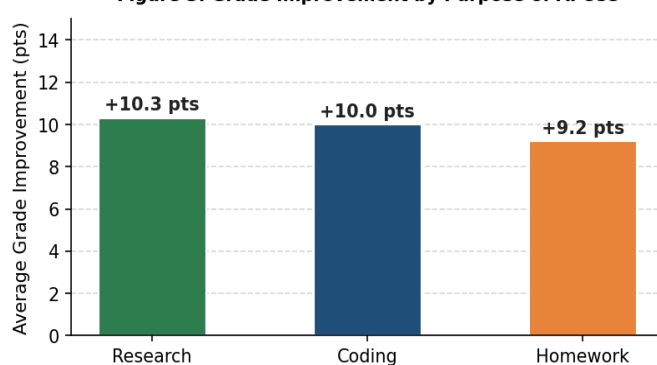


Figure 3: Grade Improvement by Purpose of AI Use

Figure 4: Study Hours vs Post-AI Grades

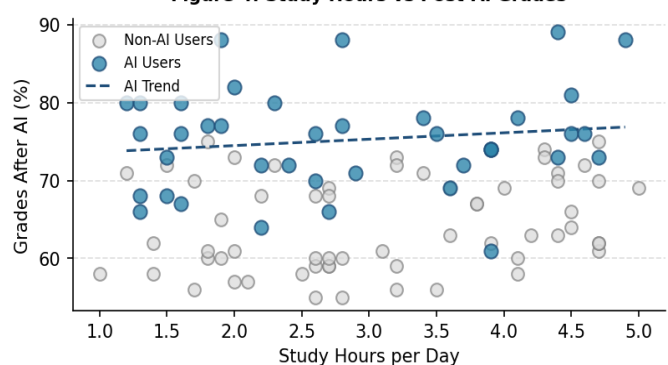


Figure 4: Study Hours vs Post-AI Grades (with trend line)

STEP 5 – INTERPRETATION LINKED TO DECISION PROBLEM

Finding	Managerial Implication
AI users: +9.8 pts; Non-AI: +0.0 pts	Strong evidence supports a policy encouraging AI adoption
All 3 tools yield ~+10 pts improvement	No need to prescribe a specific tool — student choice is effective
Research use yields highest gain (+10.3 pts)	AI especially valuable for research — integrate into library support
All purposes benefit equally (9–10 pts)	A broad policy covering all academic task types is appropriate

Table 5: Analytical Findings Linked to Managerial Decision

STEP 6 – LIMITATIONS & ANALYTICAL INTEGRITY

- Sample size:** 100 students supports descriptive analysis but limits statistical generalisation. Multi-institution samples would strengthen the policy case.
- Self-reported data:** Grades and AI usage are self-reported, introducing potential social desirability bias. Institutional records would be more reliable.
- Correlation vs causation:** Confounders such as motivation and prior ability are not fully controlled. Regression with control variables is recommended for formal policy submission.

REFERENCES

- Salamt, A. (2025). *Students' AI Usage and Academic Performance* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/aminasalamt/students-ai-usage-and-academic-performance>
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